



Security issues versus performance issues

Class 4 — Topic 1 (end), and the recap of subjects 1.1 to 1.5

420-1C1-DW Preventative Monitoring · Competency EZ01, element 1
Cybersecurity: Prevention and Intervention (LEA.D8), AEC · Dawson College, Montréal
Saturday 19 September 2026 · 2:00 – 6:00 p.m. · Online · Cyber.SoHo



Before we start: the rules that govern this material

These four rules apply to every exercise in this course, without exception.

1 Authorization decides everything

Apply what you learn only to systems you own or are authorized in writing to work on. A technique taught for defence is not lawful outside that authorization.

2 Capture only in the lab

Never capture on a shared network — home, residence or campus Wi-Fi. Capturing other people's traffic can constitute interception under section 184 of the Criminal Code.

3 Treat every capture as personal information

Logs, baselines and alert records describe the people who used those systems. Keep the minimum, store it in the lab, delete it when the exercise is marked.

4 Check every link before you open it

Every link in this course, including mine, goes through VirusTotal first. A clean result is one check among several, not a guarantee.

What we do in the next thirty minutes

Minutes	Segment	What you should be able to do afterwards
0 – 3	Why this subject exists	Say why one screen cannot separate the two categories
3 – 14	Subject 1.5, sections 1 to 5	Define both categories and run the four-question triage
14 – 20	The week-end at Atelier Nord-Ouest	Classify two real reports from evidence, not impression
20 – 27	Recap of Topic 1	Connect subjects 1.1 to 1.5 into one picture
27 – 30	Lab 1, Quiz 1 and what comes next	Know what is due on 25 September and what Quiz 1 covers

What you will be able to do at the end of today

These five sentences are the examinable outcome of this class.

- **Define** a performance issue and a security issue in words that do not overlap.
- **Explain** why one symptom can belong to either category, and sometimes to both at once.
- **Apply** a four-question method that reaches a classification from evidence rather than from impression.
- **Choose** the right verdict — performance, security, both, or undetermined — and name what each verdict obliges you to do next.
- **Connect** subjects 1.1 to 1.5 into a single picture you can use in Quiz 1, in the midterm, and in Lab 1.

1

Subject 1.5 — Security issues versus performance issues

The method that turns "it is slow" into a classification you can defend with evidence.

THE PROBLEM THIS SUBJECT SOLVES

A network that is slow and a network that is under attack look identical on the first screen you open.

A user reports a symptom, never a cause. "It is slow", "it stopped", "it is full", "it is behaving strangely".

Each of those four sentences has a routine explanation and a hostile one.

The difference is not visible in a graph of processor use. It is visible in evidence, and evidence has to be collected in a fixed order.

Five words that must mean the same thing (1/2)

Monitoring produces disagreements when two people use the same word for two different things.

E Event

Any observable occurrence in a system or on a network: a sign-in, a file written, a service restarting, a packet crossing an interface. An event is neutral. It is neither good nor bad by itself.

A Alert

A notification produced by a tool when an event or a measurement matches a rule, or crosses a threshold, that a human being configured in advance. An alert is a claim made by a tool, not a fact.

I Incident

An event, or a group of connected events, that harms or threatens confidentiality, integrity or availability, or that breaks a policy. An incident is a conclusion reached by a person after looking at evidence.

Five words that must mean the same thing (2/2)

In practice: a tool observes an event → a rule turns some events into an alert → a person decides whether the alert describes an incident.

B

Baseline — the reference database of normal activity

A written and measured description of how a system behaves when nothing is wrong, recorded over long enough to include the normal cycles of the organization: working hours, evenings, week-ends, month-end. Without a baseline, the words "high", "slow" and "unusual" have no meaning.

F

False positive and false negative

A false positive is an alert raised when nothing abnormal happened. A false negative is the absence of an alert when something abnormal did happen. Both are failures of the configuration, and both are reduced by a good baseline.

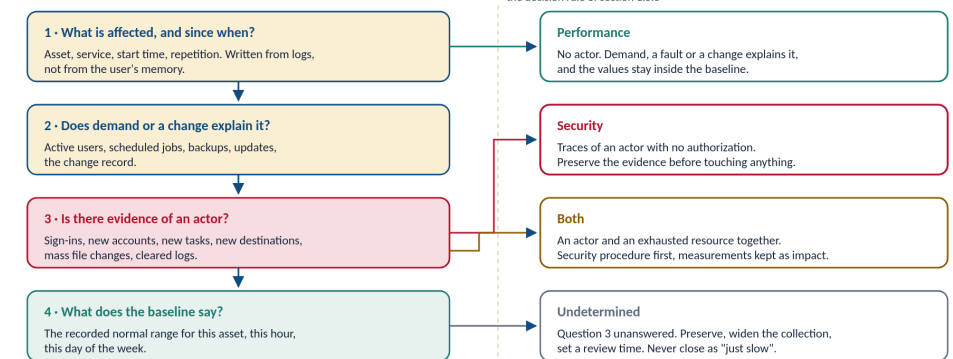
What a performance issue is

- **Definition.** A condition in which a system still behaves exactly as designed, but delivers its function too slowly, with less capacity than required, or with interruptions.
- **Because of** demand, resource limits, faults, or configuration.
- **And** no unauthorized person is involved.
- **The evidence is quantitative and repeatable.** Values rise and fall with the load.
- **It follows something** a schedule, the number of active users, or a documented change.

Security issue or performance issue: the decision path

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the decision rule of section 1.5.5



The decision path we build in the next ten minutes.



The five ordinary causes of a performance issue

Four of these five are found by reading a schedule or a change record, not by reading a graph.

1 Demand above the planned capacity

More users, larger files, a new report that runs at nine o'clock.

2 Exhaustion of a resource

CPU (Central Processing Unit) time, RAM (Random Access Memory), disk space, disk input and output, or network bandwidth.

3 A fault in hardware

A failing disk, a defective cable, a network interface negotiating at the wrong speed.

4 A change made by the organization itself

An update, a new backup schedule, a modified configuration, a new application.

5 A design limit

A single link, a single server, a service that cannot use more than one processor core.



SECTION 1.5.3

A security issue is explained by an actor — a person, a script or a program acting outside the authorization given to it.

It is a condition caused by, or indicating, an action that was not authorized, an attempt at such an action, or a weakness that makes one possible.

It affects at least one of the three properties of subject 1.2: confidentiality, integrity, availability.

The distinguishing element is not severity, and not slowness. It is the presence of an actor.



The traces an actor leaves, that a fault never leaves

Learn this list. It is the list you will look for in every lab in this course.

- **Authentication records** that match no legitimate working pattern: repeated failures, then one success, at an hour when the office is empty.
- **New objects with no change record:** accounts, group memberships, permissions, services, scheduled tasks.
- **Files created, renamed or encrypted in volume** by a single account or a single process.
- **Connections to destinations absent from the baseline**, especially outbound connections carrying volume.
- **Records that are missing:** cleared logs, gaps in a log file, a monitoring agent that stopped reporting on its own.
- **Software installed with no installation record**, or a file whose cryptographic hash no longer matches the reference value.



THE POINT THAT DECIDES MOST CASES

When you cannot explain a symptom by demand, a fault or a documented change, and traces of an actor are present, the classification is security — even when the service is still working perfectly.

A performance issue is explained by demand, faults and change.

A security issue is explained by an actor.

Write both sentences down. They are the two halves of criterion 1.4 of competency EZ01.



Why the two look alike on the first screen

Users report symptoms, not causes. Both of these facts appear in the examinations.

A security incident can produce a performance symptom

- **Encryption of a file share**, a DoS (Denial of Service) attack and mining software all consume resources.
- **The slowness is real.** It is simply not the cause.
- **If you treat the symptom**, you restart the server, lose the evidence, and the actor continues.

A security incident can produce no symptom at all

- **Copying a client folder** to a personal cloud account costs the network nothing measurable.
- **Creating a hidden administrator account** consumes no resource whatsoever.
- **This is why monitoring** cannot be limited to performance counters.



Same symptom, two possible origins

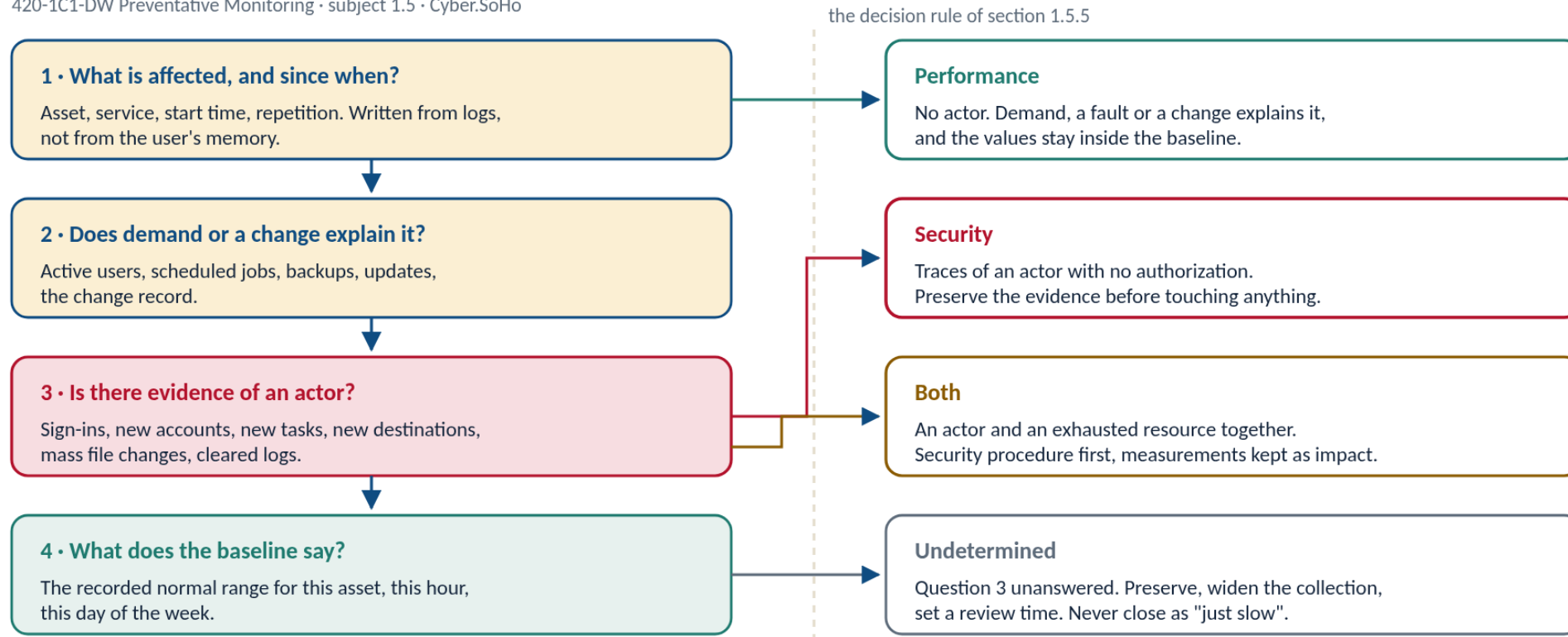
Reported symptom	Routine explanation	Hostile explanation	Evidence that decides
"It is slow"	Backup or update running; more users than planned	Encryption in progress; mining software; exfiltration	Is a job scheduled at that hour? Which process holds the disk?
"I cannot sign in"	Expired password; caps lock; directory service down	Account locked by password guessing; account disabled by an actor	How many failures, from which address, in what interval?
"The disk is full"	Log rotation failed; a large export was left behind	Staged data waiting to be exfiltrated; ransomware copies	Which folder grew, at what time, created by which account?
"Files look strange"	A synchronization conflict; a failed save	Ransomware; unauthorized modification	Do the extensions appear in the baseline? Who wrote them?
"The site is unreachable"	Link saturated; routing change; provider fault	Denial of service; interference with the network path	Where does the traffic come from, and does it resemble any user pattern?



The four-question triage, and the four verdicts

Security issue or performance issue: the decision path

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Read the left column from top to bottom in this order every time. The arrows on the right show which question produces which verdict. Question 3 is the one that separates the two categories.



Question 1 — What is affected, and since when?

Scope and time first. Everything after this depends on getting the window right.

?

What you are establishing

- The asset or the service concerned.
- The exact start time and the end time, if it ended.
- Whether it repeats, and on what cycle.
- How many systems show the same thing.

D

Where the answer comes from

- The monitoring tool's own history.
- Event logs on the affected system.
- The ticket or the message from the user — for the report only, never for the timing.
- Never from anybody's memory.



Question 2 — Does demand or a change explain it?

?

What you are establishing

- Whether an ordinary explanation already accounts for the values.
- Active users, scheduled jobs, backups, updates.
- Whether a change was made, and by whom, and when.

D

Where the answer comes from

- The job scheduler and the backup schedule.
- The change record, if the organization keeps one.
- The list of sessions open at that moment.

!

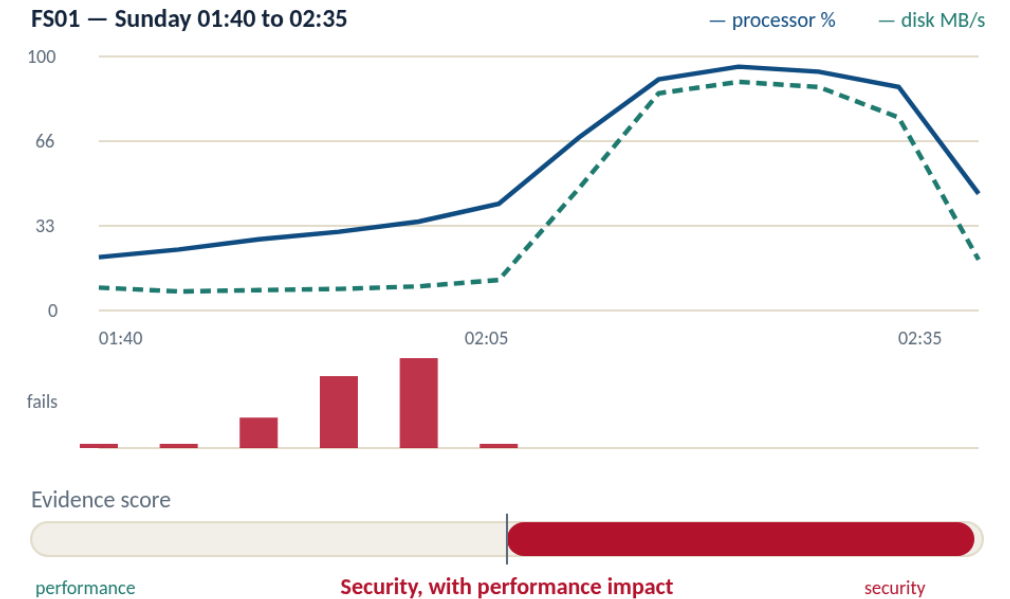
What the answer does — and does not do

- A good ordinary explanation lowers the evidence score.
- It never closes the case on its own.
- Attackers deliberately work during the backup window, because that is when nobody looks.



Question 3 — Is there evidence of an actor?

- This is the question that separates the two categories.
- **Sign-ins:** failures far above the baseline, then one success, outside working hours.
- **New objects:** accounts, permissions, services, scheduled tasks with no change record.
- **File activity:** creation, renaming or encryption in volume by one account or one process.
- **Network:** a destination that appears nowhere in the baseline, carrying volume outward.
- **Absence:** a cleared log, a gap in a log file, an agent that stopped reporting.
- **Integrity:** software with no installation record, a hash that no longer matches.



The telemetry of FS01 during the incident window: processor, disk writing, and failed sign-ins on one time axis.



Question 4 — What does the baseline say?

? What you are establishing

- How far outside the recorded normal range the values sit.
- For this asset, this hour, and this day of the week.
- Expressed as a multiple, not as an adjective.

D Where the answer comes from

- The reference database of normal activity, recorded earlier.
- The same counters, over the same length of window.
- Nowhere else — an impression is not a baseline.

= What it converts

- "It looks high" becomes "3.8 times the recorded normal".
- A number you can defend in a report, in a meeting, and in front of a client.
- Without a baseline, question 4 has no answer at all.



The decision rule — four verdicts, and only four

Apply the rule in this order. The first line that matches is your verdict.

P Performance

No trace of an actor, and demand, a fault or a documented change explains the values, and the values stay inside the baseline for this time of week.

S Security

Traces of an actor are present and no authorization explains them. Treat it as a security incident until the evidence disproves it.

B Both

Traces of an actor are present and a resource is exhausted. Handle the security path first; the resource problem is a consequence, not the cause.

U Undetermined

The evidence does not answer question 3. Preserve the data, raise the priority, escalate. Never close a case as "just slow" because no one found the cause.



WHY THE RULE IS DELIBERATELY ASYMMETRIC

Calling a security incident a performance problem leaves an actor inside the network with time to continue.

Calling a performance problem a security incident costs an escalation and some wasted hours.

The second mistake is corrected in a morning. The first one is corrected by a lawyer.

That is why doubt resolves towards security, and never towards "probably nothing".



What each verdict obliges you to do

Verdict	Immediate action	Who is informed, and what is recorded
Performance	Measure, identify the constrained resource, propose a correction with the service owner.	Service owner and the users concerned. Recorded in the monitoring record with the measurements.
Security	Preserve the evidence before touching anything: export logs, record exact times. Do not restart, do not delete.	The person named in the incident procedure. If personal information is involved, the register of confidentiality incidents applies, and Law 25 may require notifying the CAI and the persons concerned.
Both	Run the security procedure first. Keep the performance measurements as evidence of impact.	Both audiences, in the order of the security procedure. Everything recorded together.
Undetermined	Preserve the data, widen the collection, raise the priority, set a named review time.	Your supervisor, with what is missing stated explicitly. The case stays open.



Two obligations that apply to everything here

These two are not classroom etiquette. They are the terms under which you use this material.

1 Capture only where you are allowed to capture

On systems you own, or on the targets the laboratory guide designates by name — and only for the duration of the exercise. Nothing becomes a designated target because it is convenient or because it is reachable.

2 Treat every capture as personal information

Every capture, log file, baseline and alert record describes the people who used those systems. Keep the minimum, store it in the laboratory, and delete it when the exercise is marked.



Five mistakes beginners make (1/3)

- **Mistake 1 — Classifying from the severity of the symptom.** A failing disk can stop the whole studio, and it is still a performance issue. A quiet copy of one client folder costs nothing measurable, and it is a security incident. Severity measures impact, not category.
- **Mistake 2 — Stopping at the first ordinary explanation.** You find a scheduled job that could explain the load, so you close the case. Attackers work during backup windows precisely because everyone assumes the load belongs to the backup. Lower the score; keep asking.



Five mistakes beginners make (2/3)

- **Mistake 3 — Acting before preserving.** Restarting the server, deleting the suspicious files, running a cleaning tool. Each of those destroys the evidence that the classification depends on, and tells the actor that somebody noticed. Preserve first, always: export the logs, record the times, then act.
- **Mistake 4 — Calling anything unusual an attack.** The opposite error, and it costs credibility. A team that escalates every backup window stops being believed when it escalates something real. Score the evidence, name it, and let the rule decide.



Five mistakes beginners make (3/3)

- **Mistake 5 — Working without a baseline.** Without the recorded normal range, "high", "slow" and "unusual" are opinions. Question 4 cannot be answered, so every classification becomes a matter of who argues best. The correction is Lab 2: build the baseline before you need it, and record it in writing.
- **The common thread.** Every one of the five is a shortcut taken to reach a conclusion faster. The four questions exist precisely to stop that, and they cost about ten minutes.
- **What to carry away.** Evidence first, order fixed, verdict last — and write down what the evidence was, not only what you decided.



Watch: two short videos for subject 1.5

Verified on 18 September 2026. Watch them after the class, with the four questions in front of you.

- **[Indicators of Compromise — Professor Messer \(10:59\)](#)**

The concrete traces that answer question 3: account lockouts, sign-ins that could not physically have happened, unexpected resource consumption, missing log entries.

- **[Denial of Service — Professor Messer \(6:07\)](#)**

The clearest case of an attack that arrives disguised as a performance problem, including the accidental denial of service an organization causes to itself.

- **[VirusTotal — check any link before you open it](#)**

Before opening any link from any course document, including this one. A clean result is one check among several, not a guarantee. The full rule is Annex C of the Legal Notice.



2

The case — a week-end at Atelier Nord-Ouest

Two ordinary reports, the same opening sentence from the user, and two different verdicts.



Atelier Nord-Ouest — the studio and its network

Everything in this case is synthetic. No real address, host name or person appears anywhere in it.

N

The network

- RTR01 — router and firewall on a 300/100 Mbps (Megabits per second) fibre connection.
- SW01 — unmanaged switch. AP01 — wireless access point.
- FS01 — Windows file server holding client projects.
- BK01 — Linux backup server. WS01 to WS12 — twelve design workstations.
- PLT01 — large-format plotter. PRN01 — office printer.

H

How the studio works

- Fourteen employees, a design and print studio in Montréal.
- Business hours Monday to Saturday, 8 a.m. to 6 p.m.
- The backup runs every night at 01:00 and ends around 01:35.
- No staff work on Sunday.
- This is the case used for the whole integrative project.



Two reports, one week-end, one opening sentence

Both reports begin with the user saying it is slow. Only the evidence separates them.

Report A — Saturday, 13:00 to 15:00

- **The user says:** "Printing is slow, it takes forever to send a drawing to the plotter."
- **The data shows:** FS01 disk writing at 78 MB/s against a Saturday baseline of 70; the weekly archive job now starts at 13:00; five Saturdays in a row.
- **Q3 — actor evidence:** none.
- **Q4 — baseline:** inside the Saturday range.

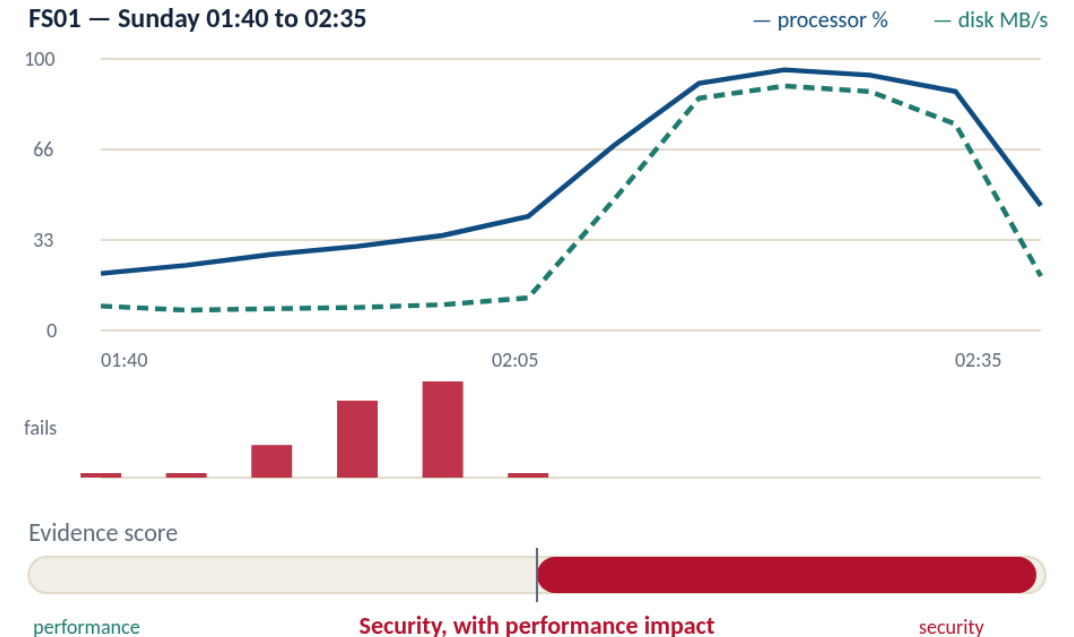
Report B — Sunday, 01:40 to 02:35

- **The user says:** "I could not sign in this morning and the server feels slow."
- **The data shows:** 412 failed sign-ins then one success at 02:05; 1 940 files with unknown extensions; a new outbound destination; the security log cleared at 02:31.
- **Q3 — actor evidence:** five separate traces.
- **Q4 — baseline:** processor 3.8×, disk 4.0×, failed sign-ins 47× the recorded normal.



Report B — what the monitoring data actually showed

- **01:40 – 01:50.** Quiet after the backup ended at 01:35, then failures begin.
- **01:55 – 02:05.** The guessing peaks at 189 failures in five minutes, then stops after one success at 02:05.
- **02:10 – 02:20.** Processor and disk rise sharply while unknown file types appear in volume.
- **02:25 – 02:35.** The activity falls away. The security log is cleared at 02:31.
- **The two lines are the symptom.** The red bars underneath are the actor.
- **Evidence score:** 13 points, from five independent traces.



Twelve measurements, five minutes apart, on FS01. Processor use in navy, disk writing in teal, failed sign-ins as bars.



Report B — the verdict, and the three obligations

v

The verdict: security, with performance impact

Traces of an actor are present, no authorization explains them, and a resource is exhausted at the same time. In the reporting this is a "Both" case: the security procedure runs first, and the processor and disk measurements are kept as evidence of impact.

1

Preserve

Export the security log, the file listing with times, and the firewall connection record — before anything else is touched.

2

Notify

The person named in the incident procedure. Not the whole office, and not the client, yet.

3

Record

Client files are involved, so the personal information rules apply and the incident goes into the register of confidentiality incidents.



WHAT YOU MUST NOT DO WITH REPORT B

Do not restart FS01, do not delete the new files, and do not run a cleaning tool.

Export the security log, the file listing with times, and the firewall connection record first.

The cleared log at 02:31 is itself evidence: it is an action, and actions have authors.

Every one of those three "helpful" actions destroys evidence that cannot be recovered.



The deliverable: the triage sheet

Row C is left empty on purpose: you fill it during Lab 1 with a third symptom of your own choosing.

Ref	Asset	Symptom and time	Q3 actor evidence	Verdict	Threshold proposed for Lab 2
A	FS01, PLT01	Printing slow, Sat 13:00–15:00	None	Performance	Alert if disk writing stays above 70 MB/s for 30 min outside 13:00–15:00
B	FS01	Lockouts and slowness, Sun 01:40–02:35	412 failures then 1 success; 1 940 unknown files; new outbound address; log cleared	Security (with performance impact)	Alert on more than 10 failed sign-ins in 5 min, and on any file extension outside the known list
C	—	Your own third symptom, chosen in Lab 1	—	—	—

The triage engine — the method written as instructions

Controls: Start, Pause on the current line for discussion, Start again to continue, Step for one line, Restart, Copy code.

- **What it is.** A self-contained HTML (HyperText Markup Language) file with JavaScript. It runs in any browser, with nothing to install, on any operating system in this class.
- **What it does.** It replays the twelve measurements from FS01, asks the four questions in order, prints the answer to each with the number that supports it, and produces the verdict.
- **Question 1** is answered by finding the first and last measurement above the baseline — that is the scope and the start time.
- **Question 2** is answered by checking the scheduled jobs and the change record for that window.
- **Question 3** adds points to an evidence score for each trace: failed sign-ins far above baseline, a success straight after them, unknown file types, a new outbound address, a cleared log.
- **Question 4** divides the highest values by the baseline, turning "it looks high" into a defensible multiple.
- **Then it refuses** to recommend a restart — because a restart destroys the evidence.



3

Recap of Topic 1 — subjects 1.1 to 1.5

One picture of the whole topic, how each subject is assessed, and what to revise before Quiz 1.



Topic 1 on one page

Topic 1 — Foundations of cybersecurity, in one page

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1.1 Why it exists

operations · money · law · trust

1.2 What we protect

Confidentiality — who may see

Integrity — what may change

Availability — when it must work

every incident report starts here

1.3 What threatens it

phishing and social engineering

malware and ransomware

denial of service

password attacks

interception on the path

insider misuse

malicious update or supplier

each one leaves a different trace

1.4 How we defend

Prevention

authentication, permissions, rules,
segmentation, updates, training

Detection ← this course

logs, monitoring tools, baselines,
alerts, integrity checking

Response

isolate, disable, block, restore, report

EZ01

element 1

criteria

1.1 · 1.2

1.3 · 1.4

Quiz 1

Midterm

Lab 1

1.5 What kind of problem am I looking at?

1 · scope and time 2 · demand or change 3 · evidence of an actor 4 · distance from the baseline

Performance

Security

Both

Undetermined

the verdict selects the procedure, the audience and what must be preserved

Keep this page beside you for Quiz 1 and for the midterm of 1 October. The lower band is subject 1.5; the three boxes above it are the vocabulary that makes the four questions answerable.



1.1 — Why an organization needs cybersecurity

Cybersecurity is not a product and not a department. It is a property of how an organization works.

O Operational

Work stops when systems stop. A studio that cannot open its project files cannot deliver.

F Financial

Interruption, recovery, replacement and lost contracts cost more than the measures that prevent them.

L Legal

In Québec, an enterprise holding personal information must protect it, keep a register of confidentiality incidents, and notify the CAI (Commission d'accès à l'information du Québec) and the persons concerned when an incident presents a risk of serious injury.

R Reputational

Clients hand over their drawings, their records and their money on the assumption that both are safe.



1.2 — The CIA triad, in the terms used for monitoring

For any event you must be able to say which property is at risk. The three are not ranked — it depends on the asset.

Property	Definition	What a failure looks like	Signals a monitoring tool can collect
Confidentiality	Only those authorized may see the information.	Data read, copied or sent to someone with no right to it.	Access to sensitive files, unusual outbound volume, access outside working hours.
Integrity	Information is changed only by those authorized, and only in ways permitted.	Files altered, encrypted or deleted; a configuration changed without a record.	File change and rename events, checksum or hash mismatches, configuration change records.
Availability	Information and services are there when they are needed.	A service that does not answer, a link saturated, a server that will not start.	Service checks, response times, error rates, resource exhaustion counters.

1.3 — Attacks and the trace each one leaves (1/2)

Attack	What it is, in one sentence	Property attacked first	Trace in monitoring data
Phishing and social engineering	A message that persuades a person to give a credential or run a file.	Confidentiality	A sign-in from an unusual place or hour; a new mail rule; a process started from an attachment.
Malware and ransomware	Software that runs without authorization; ransomware encrypts and demands payment.	Integrity, then availability	Mass file modification and renaming; new processes; unknown file extensions.
Denial of service	Traffic or requests in volume, so that a service cannot answer legitimate users.	Availability	Request rate far above the baseline, from few sources or from very many.
Password attacks	Repeated guessing, or the reuse of credentials stolen elsewhere.	Confidentiality	Failed sign-ins in volume, lockouts, then a success at an unusual hour.



1.3 — Attacks and the trace each one leaves (2/2)

Attack	What it is, in one sentence	Property attacked first	Trace in monitoring data
Interception on the path	Traffic read or altered between two parties that believe they are alone.	Confidentiality, then integrity	Certificate warnings; unexpected gateways; traffic taking an unusual route.
Insider misuse	An authorized person using their access beyond what their role permits.	Confidentiality, then integrity	Access to folders unrelated to the person's work; large copies to removable media or personal cloud.
Malicious update or supplier	Legitimate software or a supplier delivers a component that is not what it claims to be.	Integrity	A hash that does not match the published value; an unexpected outbound connection after an update.



1.4 — Prevention, detection, response

Controls are also grouped by nature: technical, administrative, operational and physical. Defence in depth places several independent controls on the same path.

P

Prevention

- Stops what it can before it happens.
- Authentication, permissions, firewall rules, network segmentation, updates, training.

D

Detection — this course

- Sees what prevention did not stop.
- Logs, monitoring tools, baselines, alerts, integrity checking.

R

Response

- Limits the damage and restores the service.
- Isolate, disable, block, restore, and report.



SUBJECT 1.5 — THE WHOLE OF TODAY IN ONE PARAGRAPH

A performance issue is explained by demand, a fault or a documented change; a security issue is explained by an actor.

The four questions establish scope and time, then demand and change, then actor evidence, then distance from the baseline.

The verdict is one of four — performance, security, both, undetermined — and it selects the procedure, the audience, and what must be preserved.

When the evidence is silent on question 3, the verdict is undetermined and the case stays open.



Turning Topic 1 into a monitoring priority list

- **The task.** Take the eight assets of the studio and state, for each one, which property of the CIA triad matters most, which attack is most likely against it, and which defence concept protects it.
- **Then calculate.** A monitoring priority from business impact and exposure — so that the ranking can be justified, not merely asserted.
- **It uses all five subjects at once:** 1.1 for the business reason, 1.2 for the property, 1.3 for the attack, 1.4 for the control, and 1.5 for the kind of issue you expect to see on that asset.
- **Deliverable.** The ranked list is section C of the monitoring plan submitted for Lab 1 on 25 September.
- **And after that.** The Tier 1 assets are the ones you instrument in Lab 2 on 3 October, and whose data you analyse in Lab 3 on 16 October.



The scoring rule — written down, and therefore repeatable

A rule that is written down can be applied by somebody else and gives the same answer. That is the difference between a ranking and an opinion.

I Impact — what the business loses

- 3 — work stops, or client information is exposed.
- 2 — work is slowed.
- 1 — the effect is minor.

E Exposure — how easily something can reach it

- 3 — reachable from outside, or by every user.
- 2 — reachable from inside the network.
- 1 — reachable only by an administrator, or physically.

× Score and tier

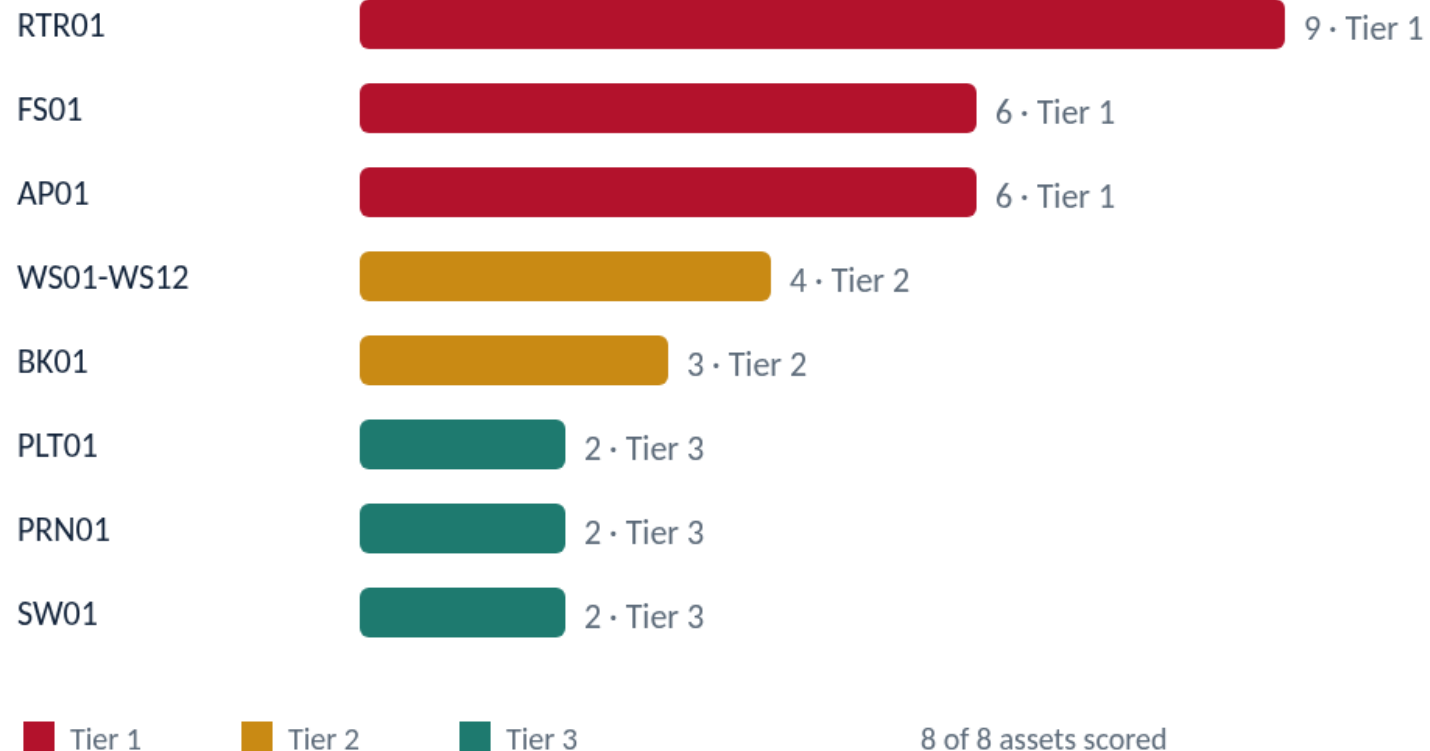
- Score = impact × exposure.
- Tier 1 when the score is 6 or more — instrument first.
- Tier 2 from 3 to 5. Tier 3 at 2 or below.



The result — eight assets, ranked and tiered

Monitoring priority — Atelier Nord-Ouest

score = business impact × exposure



RTR01 scores 9; FS01 and AP01 score 6; the twelve workstations score 4; BK01 scores 3; the plotter, the printer and the switch score 2. Three assets are Tier 1, and those are the ones instrumented first in Lab 2.



Set-MonitoringPriority.ps1 — the list produced by a script

- **Why PowerShell.** The inventory is a CSV (Comma-Separated Values) file, the studio runs Windows servers, and PowerShell reads structured data natively. The same logic would work in Python.
- **Import-Csv** turns every line of the inventory into an object whose property names come from the first line of the file.
- **Two lookup tables** translate High, Medium and Low into 3, 2 and 1 — so the words a human writes become numbers the calculation needs.
- **The loop** multiplies impact by exposure, applies the tier rule, keeps the Topic 1 columns beside the result, and prints one line per asset so the calculation can be followed while it runs.
- **Sort-Object and Export-Csv** rank the list from the highest score down and write monitoring-priority.csv beside the inventory. That file is attached to Lab 1.
- **Every line carries a comment above it.** That is the standard expected in your own submissions.



How Topic 1 is assessed

Element 1 of EZ01 is "Distinguish between the concepts linked to cybersecurity". Today completed it.

Criterion	Wording in competency EZ01	Covered in	Where it is assessed
1.1	Correct identification of issues linked to confidentiality, integrity and availability of information.	Subjects 1.1 and 1.2	Quiz 1, midterm, Lab 1 section B
1.2	Precise differentiation between various types of cyberattacks.	Subject 1.3	Quiz 1, midterm
1.3	Appropriate illustration of cyber defence concepts.	Subject 1.4	Midterm, Lab 2
1.4	Clear distinction between performance and security issues.	Subject 1.5 — today	Quiz 1, midterm, Labs 1 and 3

Lab 1 — what is due on 25 September

Everything you need was produced today. Lab 1 asks you to do it for your own network.

A Section A — the network

A map of the small network you are given, with the assets named and the connections shown.

B Section B — what must be watched

For each asset: the property of the CIA triad that matters most, and the monitoring need that follows from it.

C Section C — the priority list

The ranked list produced by impact × exposure, with the rule written down so the ranking can be checked by somebody else.

App Appendix A — the triage sheet

Today's sheet, completed with a third symptom of your own choosing in row C. It justifies which assets you ranked first.



Quiz 1 — right after this class



What it covers

All material presented up to today: subjects 1.1, 1.2, 1.3, 1.4 and 1.5. Everything in the map on the earlier slide is in scope.



What it is worth

5% of the course. It is the first of four quizzes, each worth 5%, for 20% in total.



What to have beside you

The Topic 1 map and the triage schematic. Both are downloadable from the module in SVG and PNG.



What comes next

Monday 21 September, Topic 2: the network itself — components, documentation, topology diagrams, constraints, and how monitoring needs are identified and ranked.



Where to read more (1/2) — course documents and standards

- [Cyber.SoHo — Legal Notice, Terms of Use and Acceptable Use Policy](#)

The rules that govern every course document. Section 7.2 covers monitoring and capture; Annex A is the acknowledgement each student signs before laboratory access.

- [Canadian Centre for Cyber Security](#)

The federal authority on cybersecurity in Canada: alerts, advisories, and baseline controls written for small and medium organizations.

- [NIST \(National Institute of Standards and Technology\) — Cybersecurity Framework](#)

The internationally used structure that names the functions of a security programme, including the detection function that this course belongs to.

- [MITRE ATT&CK knowledge base](#)

A catalogue of the techniques attackers use and the data sources that reveal each one — the professional version of the trace column in subject 1.3.



Where to read more (2/2) — law, and the tools ahead

- **[Act respecting the protection of personal information in the private sector \(CQLR c. P-39.1\)](#)**

The Québec statute as amended by Law 25: the register of confidentiality incidents and the notification obligations named in the verdict table.

- **[Commission d'accès à l'information du Québec](#)**

The authority to which a confidentiality incident presenting a risk of serious injury is notified.

- **[Criminal Code of Canada \(R.S.C. 1985, c. C-46\)](#)**

Sections 342.1, 342.2, 430(1.1) and 184 — unauthorized use of a computer, possession of a device, mischief in relation to computer data, and interception of private communications.

- **[Wireshark · Zabbix · Nagios · Prometheus · Grafana](#)**

The tool families named in this module and installed in Topics 3 and 4. Each is listed separately, with its own link, in the resources section of the lecture module.





Topic 1 is complete

- Subject 1.5 gave you a method: four questions, in a fixed order, answered from evidence.
- The recap connected subjects 1.1 to 1.5 into one picture, and produced the priority list for Lab 1.
- Quiz 1 follows now. Lab 1 is due Thursday 25 September.
- Monday 21 September: Topic 2 — the network to be monitored. Bring the triage sheet and the priority list.

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